

The purpose and uses of Umbrella Network

The purpose of Umbrella Network is to introduce people who are new to technology to network management and networking as a potential career. I designed it as a free tool that beginners can download and use on their own computers to gain hands-on experience with networking concepts and develop skills they can demonstrate when applying for technical jobs. I also designed the project to be able to serve as a learning and reference tool as beginners continue building their networking skills in school, personal projects, internships, and entry-level IT work.

This project is capable of gathering a computer's network information so it can diagnose potential complications by identifying local addressing and testing network connectivity, for example. I also constructed this project to include real TCP-port and ping testing while only making safe connection attempts. Another feature I added to this project assists users in constructing safe DHCP and static IP configuration plans without changing the actual network settings on their computers. I also included an Ethernet switch demonstration that shows how a switch learns MAC addresses and determines how traffic should be forwarded. Another thing I designed this project to do is support peer-to-peer networking so it can be used in scenarios where a user needs to get two programs to communicate over a network using a client-server style. Finally, this project has its own local web server made with Python code that users can run on their own computers.

These features give users hands-on experience with network troubleshooting, IP addressing, DNS resolution, TCP and ping connectivity, and network configuration concepts. The Ethernet switch and peer-to-peer portions of the project also give users experience with MAC address learning, traffic forwarding, and communication between clients and servers. The web server I designed with this project can give users experience with testing whether a web service is responding properly, working with HTTP routing and health checks, and understanding how communication between a browser and web server works.

How to use the Umbrella Network

1. Show every Umbrella Network tool: `python3 main.py --help`
2. Network diagnostics: `python3 main.py diagnose`
3. DHCP configuration plan: `python3 main.py config --mode dhcp`
4. Static configuration plan: `python3 main.py config --mode static --ip 192.168.1.25 --subnet 255.255.255.0 --gateway 192.168.1.1 --dns 8.8.8.8`
5. DNS tools - opens interactive menu: `python3 main.py dns`
6. TCP port tools - opens interactive menu: `python3 main.py ports`
7. Ethernet learning-switch demonstration: `python3 main.py switch-demo`
8. P2P server using defaults: `python3 main.py p2p-server`
9. P2P server using a chosen host/port: `python3 main.py p2p-server --host 127.0.0.1 --port 9090`
10. P2P peer example: `python3 main.py p2p-peer --name "name" (replace "name" with the user's chosen peer name) --message "Hello Umbrella"`
11. P2P peer with every connection option specified:
`python3 main.py p2p-peer --name "name" (replace "name" with the user's chosen peer name) --server-host 127.0.0.1 --server-port 9090 --message "Hello Umbrella"`
12. Web server using defaults:
`python3 main.py web`
13. Web server with host/port explicitly specified:
`python3 main.py web --host 127.0.0.1 --port 8080`

Umbrella Network Help Commands

1. `python3 main.py diagnose --help`
2. `python3 main.py config --help`
3. `python3 main.py dns --help`
4. `python3 main.py ports --help`
5. `python3 main.py switch-demo --help`
6. `python3 main.py p2p-server --help`
7. `python3 main.py p2p-peer --help`
8. `python3 main.py web --help`

Umbrella Network DNS Commands

1. Open the DNS tools: `python3 main.py dns`
2. Select hostname-to-IP lookup: 1
3. Enter a hostname when prompted: `example.com`
4. Select reverse IP-address lookup: 2
5. Enter an IP address when prompted
6. Select the default DNS test host stored in the configuration: 3
7. Exit the DNS tools: 4

Umbrella Network TCP Port Commands

1. Open the TCP port tools: `python3 main.py ports`
2. Select the option to check one TCP port on a host: 1
3. Enter the hostname or IP address when prompted: `example.com`
4. Enter one TCP port number when prompted: 80
5. Select the option to check common TCP ports: 2
6. Enter the hostname or IP address: `example.com`
7. Select the option that uses the default host/port configuration: 3
8. Exit the TCP port tools: 4

To stop the servers

1. To stop the p2p or web server: `Ctrl+C`

To open Umbrella Network:

1. Open terminal.
2. `cd ~/Folder/"Umbrella Network"` (Folder represents the name of the folder that the user has the network in.)

Sources

<https://github.com/codecrafters-io/build-your-own-x> - reference/inspiration for programming and development concepts, where applicable.

<https://chatgpt.com/> - assisted with debugging, code review, design planning, and development guidance.